Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/27/2006
PI Name:	Knobloch, Edgar Ph.D.		
Project Title:	Structure and Dynamics of Free surface flows with heating and vibration		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-7300	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vega, Jose (ETSI Aeronauticos) Martel, Carlos (ETSI Aeronauticos) Thiele, Uwe (Max Planck Intitute for Complex Systems)		
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	This project seeks to understand the effects of vibration on fluid systems. Two systems are of primary interest, the so-called Faraday system, in which a container of liquid is subjected to vertical acceleration, and the properties of drops and thin liquid films on a vibrating substrate. The focus of the project is on fundamental fluid dynamics involving free surface flows, contact line dynamics and the effects of heating and/or spatial inhomogeneities of the substrate.
Task Description:	The objective to obtain a broad understanding of the effects of vibration on pattern-forming instabilities arising in thin liquid layers as a result of Marangoni stress, wetting phenomena and contact line dynamics. These processes are of fundamental importance in the flow of liquids under microgravity conditions. Under these conditions Marangoni effects can be used to promote lubrication, coating and/or cooling but these processes are associated with instabilities that may be reduced or suppressed (and in some cases enhanced) by residual g-jitter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding achieved will be of fundamental importance for related phenomena on the earth, in particular in the currently exciting area of nanofluidics.
Task Progress:	This project seeks to understand the effects of vibration on fluid systems. Two systems are of primary interest, the so-called Faraday system, in which a container of liquid is subjected to vertical acceleration, and the properties of drops and thin liquid films on a vibrating substrate. The focus of the project is on fundamental fluid dynamics involving free surface flows, contact line dynamics and the effects of heating and/or spatial inhomogeneities of the substrate, with a view to obtaining a broad understanding of the effects of vibration on pattern-forming instabilities arising in thin liquid layers as a result of Marangoni stress, wetting phenomena and contact line dynamics. These processes are of fundamental importance in the flow of liquids under microgravity conditions where Marangoni effects can be used to promote lubrication, coating and/or cooling, but are associated with instabilities that may be reduced or suppressed (and in some cases enhanced) by residual g-jitter. These phenomena are basic to nanofluidics as well. In the year 11/1/04-10/31/05 the PI has pursued two main themes: (i) The dynamics of the Faraday system, i.e., the properties of surface gravity-capillary waves in a vertically vibrating container of liquid. When the amplitude of vibration exceeds a critical value the flat surface becomes unstable to standing waves, which in turn are subject to a variety of secondary instabilities at finite amplitude. In low viscosity liquids the most important of these are associated with the excitation of a streaming flow driven through Reynolds stresses generated in oscillatory viscous boundary layers at rigid walls and the free surface. This streaming flow in turn interacts with the waves responsible for the oscillatory boundary layers. Thus in this regime the surface waves are described by a set of coupled amplitude-streaming flow equations. The resulting system constitutes a new class of pattern-forming equations. (ii) The structure and stability properties of thin liquid films or drops on a horizontal substrate, subject to normal vibration, Marangoni stresses and surface heterogeneity, and the dynamics of such films and drops on an inclined substrate. Progress Report: (i) The PI discovered that the interaction between Faraday waves and the associated streaming flow can lead to remarkable relaxation oscillations in this system, whose origin was traced to the disparity in time scales between the (slow) decay time of free surface gravity-capillary waves, and the (even slower) decay time of the streaming flow, that is characteristic of low viscosity liquids. The oscillations can be interpreted in terms of slow drifts along manifolds of equilibria or periodic orbits with fast jumps between them triggered by local or global bifurcations. The resulting relaxation oscillations are important because in appropriate circumstances they can be the first state that is observed as the amplitude of the parametric forcing increases and the flat surface loses stability. Details can be found in Higuera, Knobloch and Vega, Physica D 201, 83-120 (2005). The coupled amplitude-streaming flow equations have been simulated in two dimensions with periodic boundary conditions in the horizontal. These simulations reveal a variety of secondary bifurcations leading to drift that are present due to the coupling to the streaming flow. However, the details turn out to be sensitive to surface contamination (Martin and Vega, J. Fluid Mech., in press). A phenomenological model of weakly damped Faraday waves and the associated mean flow suitable for three-dimensional studies has been developed and used to study the range of stable standing waves in wavenumber space and the role of the mean flow in generating various secondary instabilities that limit this range (Vega, Ruediger and Vinals, Phys. Rev. E 70, 046306, 2004). The PI together with Dr U Thiele and Prof Vega have derived a fully nonlinear amplitude equation for the thickness of a thin liquid film on a heated rapidly vibrating substrate, focusing on the case of normal vibration. The resulting equation has been used to elucidate the change in drop shape and internal circulation due to vibration for both horizontal and inclined substrates. Overall, vibration was found to stabilize the thin film by delaying (secondary) instabilities and flattening drop shapes (Thiele, Vega and Knobloch, J. Fluid Mech., in press). The PI together with Dr Thiele has examined the effect of pinning of drops due to surface microstructure with a view to elucidating the process by which drops begin to slide downslope as the slope increases. The dominant surface heterogeneity was taken to be in the disjoining pressure that describes the wetting properties of the surface in response to the molecular interactions between the liquid and substrate. The results indicate that the depinning process typically occurs via the so-called 'sniper' bifurcation, and the pinned states and their stability properties have been characterized. Together with Prof A Bergeon the PI has initiated a study via direct numerical simulation of the thin film equation on a heated inclined substrate, focusing on parameter regimes where earlier work of Dr Thiele and the PI indicates the absence of stable waves of constant form. Preliminary results indicate the presence of a variety of time-dependent states resembling interacting solitary waves, as well as the possibility of film rupture. Dr A Yochelis (postdoctoral researcher at Berkeley) has recently joined the project and is working on the influence of

wetting and heat exchange on droplets with surface melting and/or freezing. This problem is of great interest since it includes changes in the morphology of the substrate in response to solid deposition due to freezing, and therefore involves coupled droplet motion and simultaneous substrate evolution.

In the year 11/1/05-9/3/06 the PI has added a third line of investigation:

(iii) Together with Mr J Burke (graduate student researcher at Berkeley) the PI has been studying spatially localized states such as drops and fronts with a view to understanding their origin and properties. He has been able to relate to identify so-called {\it pinning} regions in parameter space with an infinite multiplicity of stable localized states with different morphology, and has provided a simple and intuitive explanation for this remarkable behavior within several different physical systems. The simplest problem exhibiting this type of behavior is described by the bistable Swift-Hohenberg equation on an unbounded domain and related equations. This system is variational in time and Hamiltonian in space. These two properties aid greatly in understanding of its behavior, both within the pinning region and outside it, because of the presence of a Maxwell point at which the energy, suitably defined, of the spatially periodic state equals that of the trivial state. Localized states biasymptotic to this state can be viewed as bound states of pairs of fronts bounding the state at either end; owing to locking between the fronts and the spatially periodic structure between the fronts the Maxwell point is broadened into a pinning region; inside this region localized states of opposite parity lie on branches that 'snake' back and forth between the boundaries of the pinning region, forming a snakes-and-ladders structure Outside this region the system evolves to the lower energy state: the fronts unpin and the lower energy state invades the higher energy state, eroding it at a rate that can be computed theoretically. These and other developments are reported in a series of three papers by Burke and the PI. The stability properties of these states in two spatial dimensions are being investigated; related fully two-dimensional localized states are also present.

Related behavior is found in nonvariational systems, and the PI has identified several convection systems where localized convection in the form of {\it convectons} is present, and exhibits related properties. This work is being done with Dr O Batiste (Barcelona) and Prof A Bergeon (Toulouse).

Parametrically driven systems provide another system where localized states called oscillons are present. The PI, together with Burke and Yochelis has undertaken a detailed study of localized structures in the vicinity of the 2:1 resonance tongue. It is significant that in this system bistability is generated by the parametric forcing itself; the parameter region where this occurs, near one of the tongue boundaries, is where localized states are found. The team has identified several distinct types of localized structures, in addition to the original oscillons. These take the form of holes in a homogeneous oscillating background, or reciprocal oscillons which can be viewed as holes in a backgound of spatially periodic oscillations. There is a similar wealth of front-like structures connecting out-of-phase states. All of these are organized by Maxwell-like points in parameter space as explained in detail in two preprints. In two spatial dimensions the corresponding states (spots etc) exhibit similar properties, and are stable for comparable parameter values. This work is preliminary to a parallel study of drops on a heated and/or vibrating substrate.

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